



UNIVERSITY OF KELANIYA – SRI LANKA
FACULTY OF SCIENCE

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Computer Science
COSC 44263 – Theory of Computing

No of Questions: Four (04)

No. of pages: Three (03)
Answer All Questions

Time: Two and a Half (2 1/2) hrs

Question 1

- (a) The formal description of the Finite Automaton (FA) M_1 is $(\{1,2,3\}, \{a,b\}, \delta, 1, \{2\})$, where δ is given by the following Table 1.

Table 1. Transition Table of M_1 .

	a	b	ϵ
1	{3}	-	{2}
2	{1}	-	-
3	{2}	{2,3}	-

- i. Draw the state transition diagram for the FA M_1 .
- ii. Is this FA Deterministic or Nondeterministic? Justify your answer.
- iii. Does the machine M_1 accept the input ϵ ?
- iv. Produce two sample strings that can be accepted by this machine.
- v. Write the language accepted by the M_1 using set notation.

- (b) Using the nondeterminism concept proves that the class of regular languages is closed under the union operation.

- (c) i. Give a low-level description for the Nondeterministic Finite Automata (NFA) M_2 recognizing the language $(01 \cup 001 \cup 010)^*$.
- ii. Draw the state transition diagram for machine M_2 .
 - iii. Convert this NFA to an equivalent Deterministic Finite Automaton (DFA). Give only the portion of the DFA that is reachable from the start state.

Question 2

- (a) Consider the Pushdown Automaton (PDA) G_1 in Figure 2. G_1 recognizes the language $\{ww^R \mid w = (a+b)^*, R \geq 0\}$.

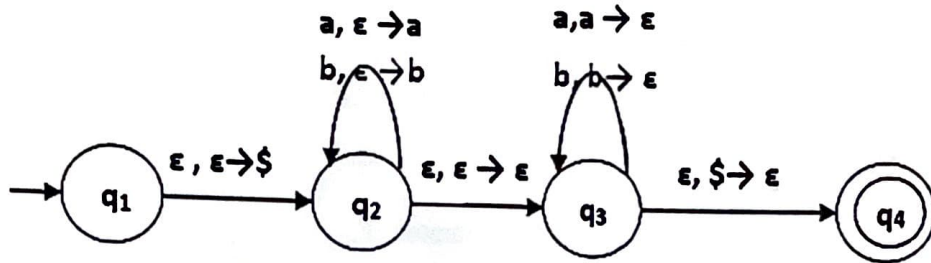


Figure 2. Pushdown Automaton G_1

- i. Write the low-level formal description for the G_1 .
 - ii.
 - A. Write an example notation from the above diagram that you have used to signify a stack reading while producing a transition.
 - B. What is/are the state/s involved in this transition?
- (b) Consider the following Context Free Grammar (CFG) G_2 :
- $$\begin{aligned}
 S &\rightarrow aXbX \\
 X &\rightarrow aY \mid bY \mid \epsilon \\
 Y &\rightarrow X \mid c
 \end{aligned}$$
- i. What are the variables of G_2 ?
 - ii. What are the terminals of G_2 ?
 - iii. What is the start variable of G_2 ?
 - iv. Clearly stating each step convert this CFG into an equivalent CFG in Chomsky Normal Form.
- (c) Give context-free grammar that generates each of the following languages.
- i. All binary strings with both an even number of zeroes and an even number of ones.
 - ii. All strings of the form $0^a 1^b 0^c$ where $a + c = b$. (Hint: it's the concatenation of two simpler languages.)

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Question 3

- (a) i. Write down the formal definition of a Turing Machine.
ii. By examining the above formal definition of a Turing machine, answer the following questions and justify your answer.
- A. Can a Turing machine ever write the blank symbol $\sqcup$ on its tape?
 - B. Can the tape alphabet Γ be the same as the input alphabet Σ ?
 - C. Can a Turing machine's head ever be in the same location in two successive steps?
 - D. Can a Turing machine contain just a single state?
- (b) i. Define the Countable Set.
ii. Let R be the set of real numbers. Prove that the set R is uncountable.
iii. What is the method introduced with the above (b) ii. Proof?
- (c) Prove the theorem "Every multitape Turing Machine has an equivalent single-tape Turing Machine."
- (d) Prove the theorem "A language is Turing decidable if and only if it is Turing-recognizable and co-Turing-recognizable."

Question 4

- (a) Consider the following statement.
"The choice of computational model can affect the time complexity of languages"
Is this statement true or false? Justify your answer.
- (b) Let $PATH = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph that has a directed path from } s \text{ to } t \}$. Prove that $PATH \in P$.
- (c) Describe the problem of RELPRIME by producing an example.
- (d) i. Briefly describe the problem of SUBSET-SUM problem by producing an example.
ii. What is the proof idea to prove the *SUBSET_SUM* is in NP?
- (e) i. Provide the definition for the NP-completeness.
ii. Give one example of an NP-complete problem.

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